

An electrochemical–thermal model of lithium-ion battery and state of health estimation

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ABSTRACT

The performance of lithium-ion batteries deteriorates with the passage of time, estimating State-of-Health of batteries correctly is important. In this paper, an electrochemical-thermal model including the electric double layer model is developed. In the frequency domain, sensitivity analysis of the electrochemical processes during battery aging are carried out based on electrochemical impedance spectroscopy and distribution of relaxation time method. In the time domain, single parameter sensitivity analyses are performed. Methods of obtaining different types of parameters in the model are studied, and the electrochemical parameter values of the battery under different cycles are obtained. Finally, some key parameters with higher correlation with State-of-Health are selected to achieve State-of-Health estimation of the battery.

1. Introduction

Lithium-ion traction battery is one of the most important energy storage systems for electric vehicles [1,2], but batteries will experience the degradation of performance (such as capacity degradation, internal resistance increase, etc.) in operation and even cause some accidents because of some severe failure forms [3–5]. To ensure a pleasant and safe driving experience, the estimation of State-of-Health (SOH) is crucial for the electric vehicle application [6].

The SOH estimation methods are mainly divided into two categories: data-driven method and model-driven model [7,8]. Data-driven model directly extracts some aging features from various SOH-related experimental data, such as the incremental capacity (IC) curve [9–12], Electrochemical Impedance Spectroscopy (EIS) [13,14], temperature characteristics [15]. IC Analysis is one of the most common methods to estimate SOH, because IC curves are easily obtained from constant current charging or discharging tests which are very common in aging experiments of lithium ion batteries. The shape features of IC curves (such as the height of peaks, the x-coordinate of peaks and the area of peaks) are often extracted to estimate SOH [10,11]. However, noise is easily introduced during the acquisition of IC curves, causing that some features is difficult to be extract. Hence, some researches reconstruct IC curves combining a voltage model [16,17] or develop the model containing the feature of IC curves [18] to avoid this problem. Apart from IC curve, EIS measurements are also highly sensitive to noise and even has

more strict requirements to test equipment. In general, the achievements of high accuracy of data-driven method is heavily dependent on the accurate acquisition of raw data. Besides, most of data-driven methods do not focus on the aging mechanism explanation, which becomes another limitation of data-driven method.

To explain the aging mechanism, some model-driven methods developed electrochemical models, which describe the electrochemical processes inside batteries in detail [19–24]. The full-order electrochemical model consists of many nonlinear partial differential equations and a series of electrochemical parameters, which makes it too complex to achieve practical application. Hence, many researchers focus on the simplification of the electrochemical model. Zou C et al. proposed a framework for the model simplification and discussed many assumptions which can be introduced in simplified models [23]. Our model in this paper also introduces some assumptions which are shown in Section 3.

In general, the SOH estimation based on electrochemical models is achieved by investigating the relationship between SOH and model parameters. Rui X et al. extracted five characteristic parameters which are highly related to the battery degradation from an electrochemical model to achieve SOH estimation [19]. Lyu C et al. summarized ten model parameters which shows strong correlations with the battery degradation [20]. Hence, the high-accuracy and high-efficiency acquisition of model parameters is crucial for SOH estimation. The methods to obtain model parameters can be divided into two types, namely, direct

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measurements and parameter identifications. Some geometric parameters can be measured by disassembling batteries [21], while many electrochemical parameters need to be obtained by parameter identification methods. Dokko K et al. developed an equivalent circuit model and estimate the kinetic parameters based on EIS data [25]. Forman J C et al. chose the genetic algorithm to identify the electrochemical parameter of Pseudo-2-dimensional (P2D) model combining with seven experimental cycling data [22]. Li Y et al. achieved the estimation of internal battery parameters at a low computational cost based on a reduced-order battery model and the constrained ensemble Kalman filter [24].

This paper aims to develop an electrochemical model to achieve the accurate simulation and SOH estimation in whole life cycle of batteries. At first, EIS data in cycling aging experiments is analyzed to provide the guidance to modeling. Then an electrochemical-thermal model is developed based on P2D model. In this model, the electric double layer is described more detailed than P2D model to adapt its large change in aging processes and variations of some parameters with temperature are described by a concentrated mass thermal model to compensate simulation errors caused by temperature change. Regarding the parameter acquisition, all parameters are divided into three groups, which correspond to different acquisition methods. Finally, seven highly SOH-related parameters are determined, and SOH estimation is achieved using neural network.

2. Experiment and sensitivity analysis

2.1. Aging experiment

To research the aging behaviors of lithium ion batteries, the accelerated cycling aging experiment is carried out. We choose Panasonic NCR18650BD lithium ion battery as our experimental object. The experimental platform and information interaction used in this paper are shown in Fig. 1(a). The charging and discharging device is used to adjust SOC and apply specific current excitation on batteries. The incubator is used to simulate the environment temperature during the experiment. The temperature recorder is used to record the surface temperature of batteries. The electrochemical workstation is used to obtain the EIS of batteries

A summary of the types of devices used in this experiment is shown in Table 1.

The accelerated cycling aging experiments include constant current charging and discharging aging tests and performance reference tests (PRTs). Based on the recommend operation range of NCR18650BD lithium ion battery, the charge rate and the discharge rate are set as 0.5C and 3C, respectively, and the charging end voltage and the discharging end voltage are set as 4.2 V and 2.5 V, respectively. In order to accelerate the battery aging process, the ambient temperature is set to 40 °C. The temperature reduces to 25 °C, when PRTs are carried out after every certain number of cycles. PRTs include the EIS measurement, the Multi-Stage Constant Current (MSCC) condition test and the China light-duty vehicle test cycle-passenger (CLTC-P) condition test. Both aging tests and RPTs are carried out strictly in accordance with the experimental procedures shown in Fig. 1(b). The EIS measurement, MSCC operating condition test, and CLTC-P condition test are inserted in the entire aging process.

2.2. Sensitivity analysis

The aim of sensitivity analysis is to find out which electrochemical process is sensitive to the battery degradation and needs to be detailedly described in the electrochemical model. In this Section, we analyze the sensitivity of different electrochemical processes based on EIS and the distribution of relaxation time (DRT). EIS is currently one of the most commonly used electrochemical detection methods [13,14,26]. And DRT, which is based on EIS data, can achieve better decoupling effect

than EIS [27–30]. We summarize the common characteristics of all EIS data in aging experiments as shown in Fig. 2. The Nyquist curve consists of two semi-circles and one straight line and three characteristic points are marked using green circles. From left to right of Fig. 2, the first characteristic point is the point which has zero imaginary part and it is regarded as the Ohmic resistance. The frequency range of this points is 400–600 Hz according to all measured EISs in aging experiments. The second characteristic point is the intersection of the first semi-circle and the second semi-circle and its frequency range is 10–100 Hz. Regarding the third characteristic point, it is the intersection of the second semi-circle and the straight line and its frequency range is 0.1–1 Hz. Based on above analysis, we can divide the Nyquist curve into five parts: extremely high frequency region, Ohmic resistance, high frequency region, intermediate frequency region, and low frequency region. In the extremely high frequency region, Nyquist curve shows an inductance characteristic caused by the skin effect of test wires. In the high frequency region, Nyquist curve is dominant by SEI impedance and the contact impedance between active materials and collectors. The impedance caused by the charge transfer and charge accumulation in the electric double layer is mainly shown in the intermediate frequency region. And the low frequency region shows the solid phase and liquid phase diffusion impedance.

The Nyquist diagrams from 0 to 220 cycles in aging experiments are shown in Fig. 3. To make a clear comparison, all Nyquist curves are moved horizontally to make their Ohmic resistance as one point. To ensure that the EIS change can be fully observed, the six SOC points include 5% SOC, 15% SOC, 35% SOC, 55% SOC, 75% SOC, and 95% SOC are selected for EIS measurements with a frequency range of 0.01–1000 Hz every certain number of cycles.

It can be seen from Fig. 3 that the first semi-circle in the high frequency region has no obvious change. As the number of cycles increase, the size of the second semi-circle in intermediate frequency region increases dramatically in total, indicating that the electric double layer impedance is highly sensitive to aging. However, it can be noticed that the low frequency part of Nyquist curve is more and more overlapped by the second semi-circle of the intermediate frequency region, which is more obvious in Fig. 3(a)(b)(f), so the change of EIS in the low frequency region is difficult to be observed only based on EIS data.

To accurately decouple the internal processes of the lithium-ion battery, DRT method is introduced. This method expresses the entire impedance of the battery as:

$$Z(\omega) = R_{Ohm} + R_{pol} \int_0^{\infty} \frac{\gamma(\omega)}{1 + j\omega\tau} d\tau, \quad \int_0^{\infty} \gamma(\omega) d\tau = 1 \quad (1)$$

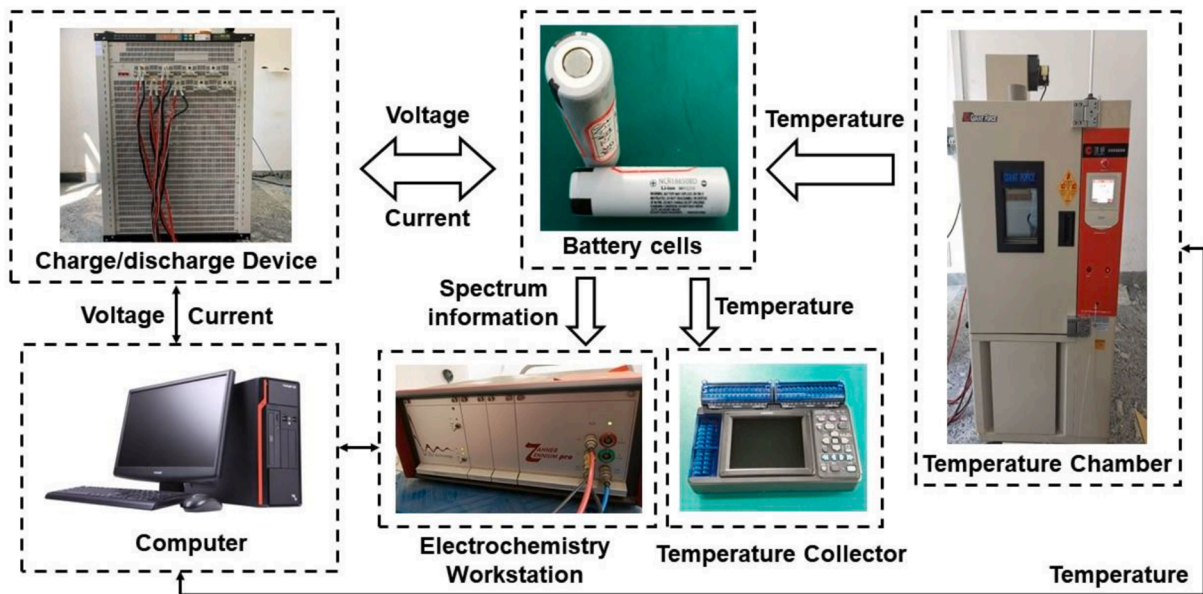
where $\gamma(\omega)$ is relaxation time distribution function, R_{Ohm} is Ohmic resistance (Ω), R_{pol} is polarized resistance (Ω) and τ is the time constant (s^{-1}).

As shown in Eq. (1), DRT method describes the impedance of batteries using RC model. The impedance of one RC element (the parallel connection of one resistance and one capacitance) is described by $\frac{R}{1 + j\omega\tau}$,

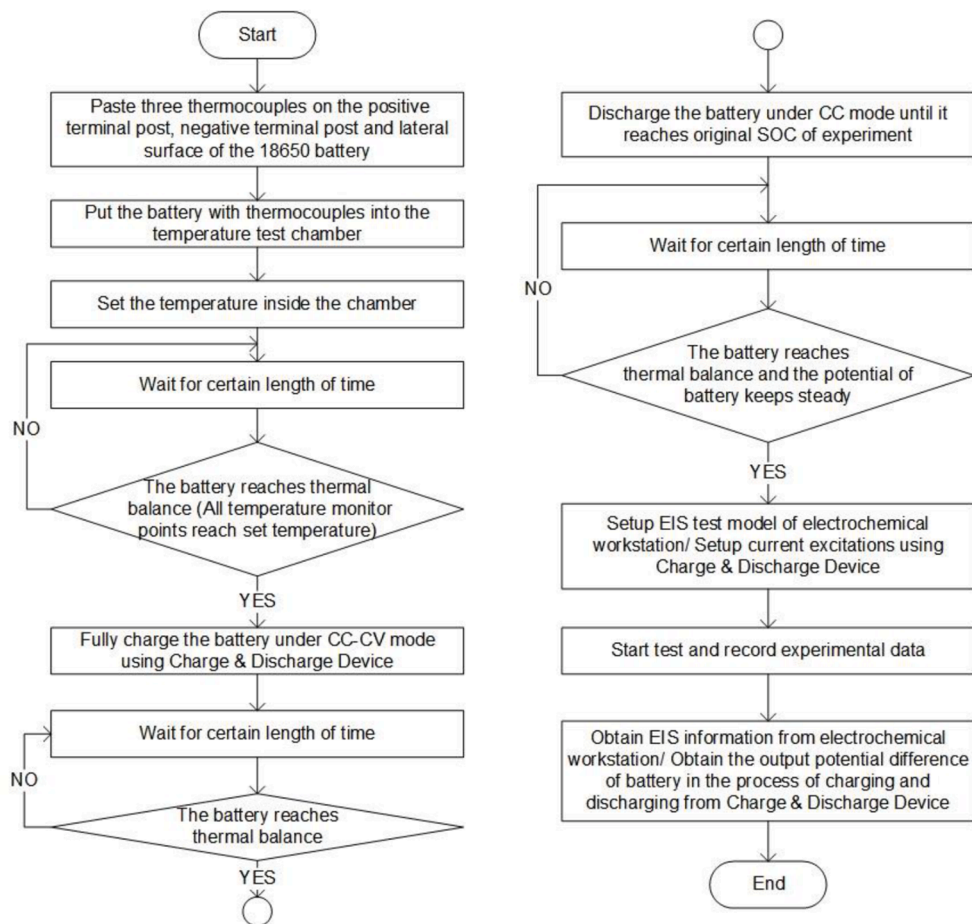
so $R_{pol} \int_0^{\infty} \frac{\gamma(\omega)}{1 + j\omega\tau} d\tau$ can be regarded as the sum of all impedance of RC

elements in the RC model. R_{Ohm} , R_{pol} and $\gamma(\omega)$ can be solved based on EIS data. The measured real part and imaginary part of impedance in specific frequency points can be put into Eq. (1) and an equation set about measured impedance values in all measured frequency points can be obtained. R_{Ohm} , R_{pol} and $\gamma(\omega)$ are solved by solving the equation set and the DRT diagram can be drawn using $\gamma(\omega)$ as vertical axis and $\lg(\omega)$ as horizontal axis. In the DRT diagram, one peak corresponds to processes with similar relaxation times.

Since different electrochemical processes inside the battery have specific relaxation time ranges, we can decouple the reaction processes inside the battery based on DRT diagram [28,29]. The DRT diagram



(a)



(b)

Fig. 1. (a) The experimental platform and information interaction (b) Experiment steps.

Table 1
Types of devices.

Device	Type
Electrochemistry Workstation	Zennium pro
Charge/discharge Device	CT-8008-5V100A-NTFA
Temperature Chamber	ECT-225-40-CP-AR
Temperature Collector	LR8400

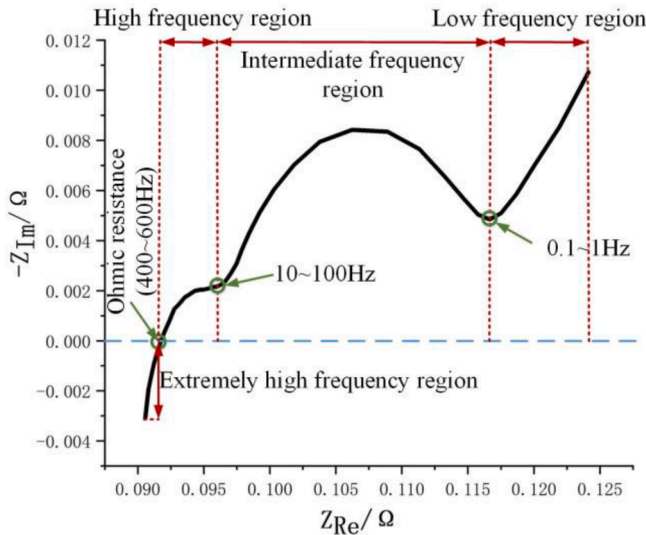


Fig. 2. Region division of Nyquist diagram of measured EIS.

corresponding to the EIS under different SOCs is shown in Fig. 4. The DRT curves can be divided into four parts from left to right according to the range of the relaxation time, and each part has an independent peak, indicates that the four sections correspond to separate electrochemical processes.

Based on some researches about DRT analysis, different electrochemical processes have their relatively constant relaxation time ranges [27–30]. Four parts shown in Fig. 4 are also divided referring to previous researches, so it can be deduced which electrochemical process is corresponded to by a specific part. S1 peak and S2 peak correspond to the solid and liquid phase diffusion process and the electric double layer effect and the charge transfer process, respectively. Even though S1 peak overlaps with S2 peak in some extent in the latter aging period, it can be noticed that the height of S1 peak shows an upward trend, which indicate that the solid and liquid phase diffusion process also shows high sensitivity to aging. Hence, the impedance of the solid and liquid phase diffusion process also increases with the increase of the cycle number, which can be not included by analyzing EIS. The peak height of the S2 peak becomes higher and shifts to the left with the increase of the cycle number, which is consistent with the increase of the second semi-circle in EIS diagram. The S3 peak and S4 peak respectively represent the diffusion process of lithium ions in the SEI film and contact impedance, which keep relatively steady and correspond to the first semi-circle change in aging process.

Based on above analysis, we can conclude that processes sensitive to battery degradation include the solid and liquid phase diffusion process and the electric double layer effect and the charge transfer process, which need to be described in detail in electrochemical battery degradation model. Regarding the diffusion process of lithium ions in the SEI film and contact impedance, even though they have capacitance characteristic because they show peaks in DRT diagram, but they can be regarded as pure resistance process because the excitation frequency in general time domain applications does not reach such high that these processes can show capacitance characteristic in the excitation

frequency. Hence, the impedance of these two processes are merged into the ohmic polarization impedance in the following sections.

3. Electrochemical-thermal model

The overview of the electrochemical model is shown in Table 2 and the model parameters are shown in Table 3 and the detailed description of this model is given as the following sections.

3.1. Electrochemical model

Parts of the electrochemical-thermal model (from Eq. (2) to Eq. (8)) is developed based on P2D model. P2D model divides the lithium-ion battery into positive and negative electrode current collectors, positive and negative electrodes, and separator. Based on porous electrode theory and concentrated solution theory, the process of insertion and extraction of lithium-ions on the electrode, the solid phase diffusion and liquid phase diffusion process of lithium-ions, the charge transfer reaction process at the interface of solid and liquid phase, the solid and liquid phase Ohm's law, and law of mass conservation are described in detail [31,32]. Among above processes described by P2D model, the description of the charge transfer reaction process at the interface of solid and liquid phase is replaced by more detailed electric double layer model in proposed electrochemical-thermal model as Eq. (9)–Eq. (12). Referring to P2D model, some assumptions are introduced into the electrochemical model:

- (1) Active particle in two electrodes are assumed as uniform spherical particles.
- (2) The battery working process is simplified to solid and liquid phase process.
- (3) Lithium-ions only move inside the battery through diffusion and migration, without considering convection.
- (4) All side effects are ignored.
- (5) The current collector is assumed as an infinite conductor, and only electrochemical processes along the thickness of the electrode are considered.
- (6) The electric double layer structure at the solid-liquid phase interface conforms to the GCS (Gouy-Chapman-Stern) model.
- (7) Lithium ion fluxes of the positive and negative electrodes are assumed as evenly distributed.
- (8) The liquid phase lithium ion concentration distribution curve in the separator is a straight line.

Eq. 2 in Table 2 is basic equation which describes the composition of battery terminal voltage. In Eq. 2, η_{con} can be obtained by solving Eq. (3), Eq. (6) and Eq. (7). As above assumptions, lithium-ion fluxes of the positive and negative electrodes are assumed as evenly distributed, so the average lithium-ion fluxes of the positive and negative electrodes as shown in Table 2 can be obtained as

$$j_n \approx \frac{I(t)}{a_n F l_n A_{0,n}} \quad (18)$$

$$j_p \approx \frac{I(t)}{a_p F l_p A_{0,p}} \quad (19)$$

where $I(t)$ is the external current (A), Subscripts n and p respectively represent negative electrodes and positive electrodes, a is the specific surface area of active particles (m^{-1}), l is the electrode thickness (m), A_0 is the effective area of electrode (m^2).

The derivation process of the electric double layer model is as follows:

Two different description of the electric double layer model are proposed in previous researches. One description is from P2D model, the researchers used the Butler-Volmer (B-V) equation to directly calculate

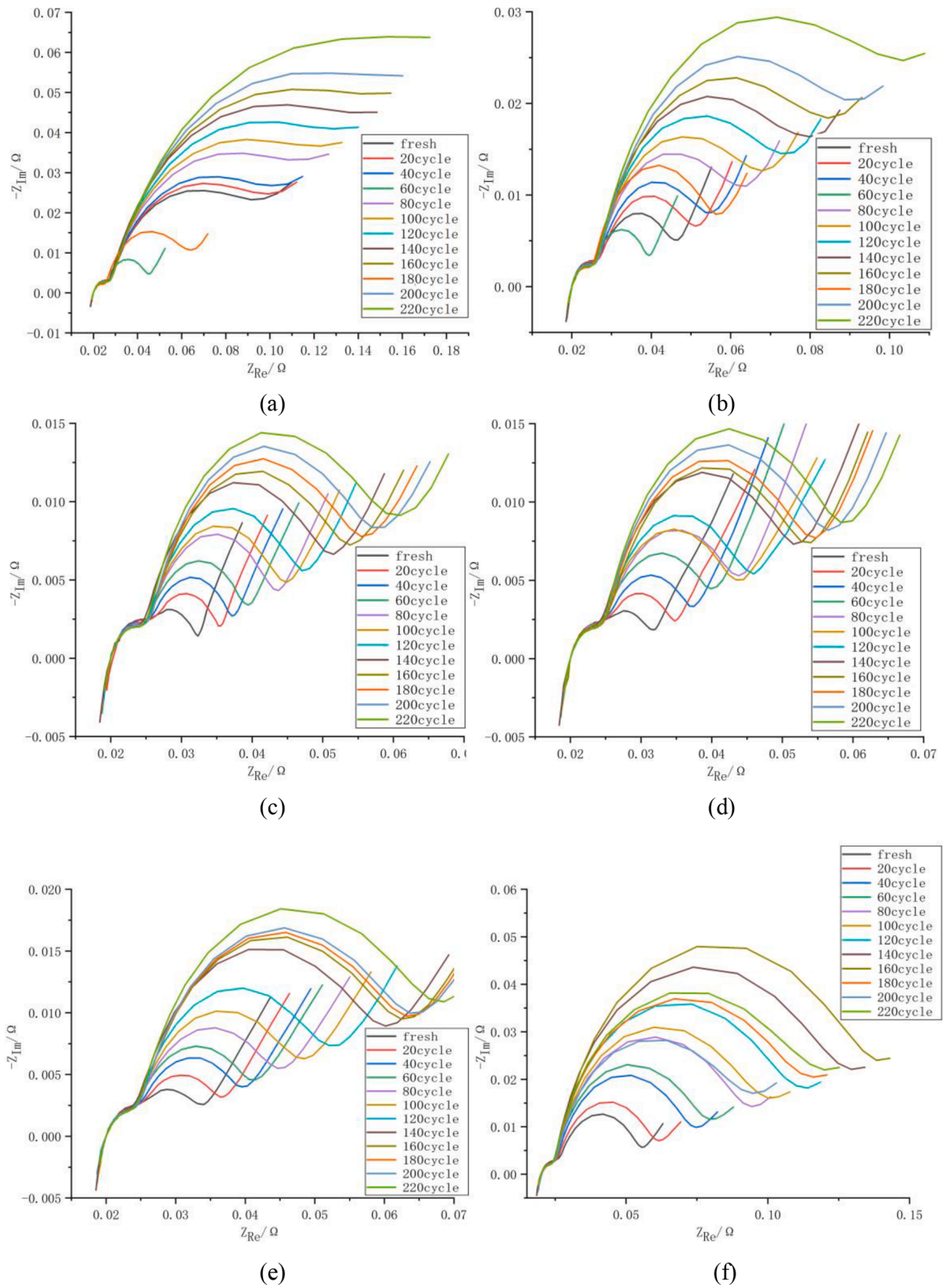


Fig. 3. Nyquist diagrams of different cycles under different SOC (a) 5%SOC (b) 15%SOC (c) 35%SOC (d) 55%SOC (e) 75%SOC (f) 95%SOC.

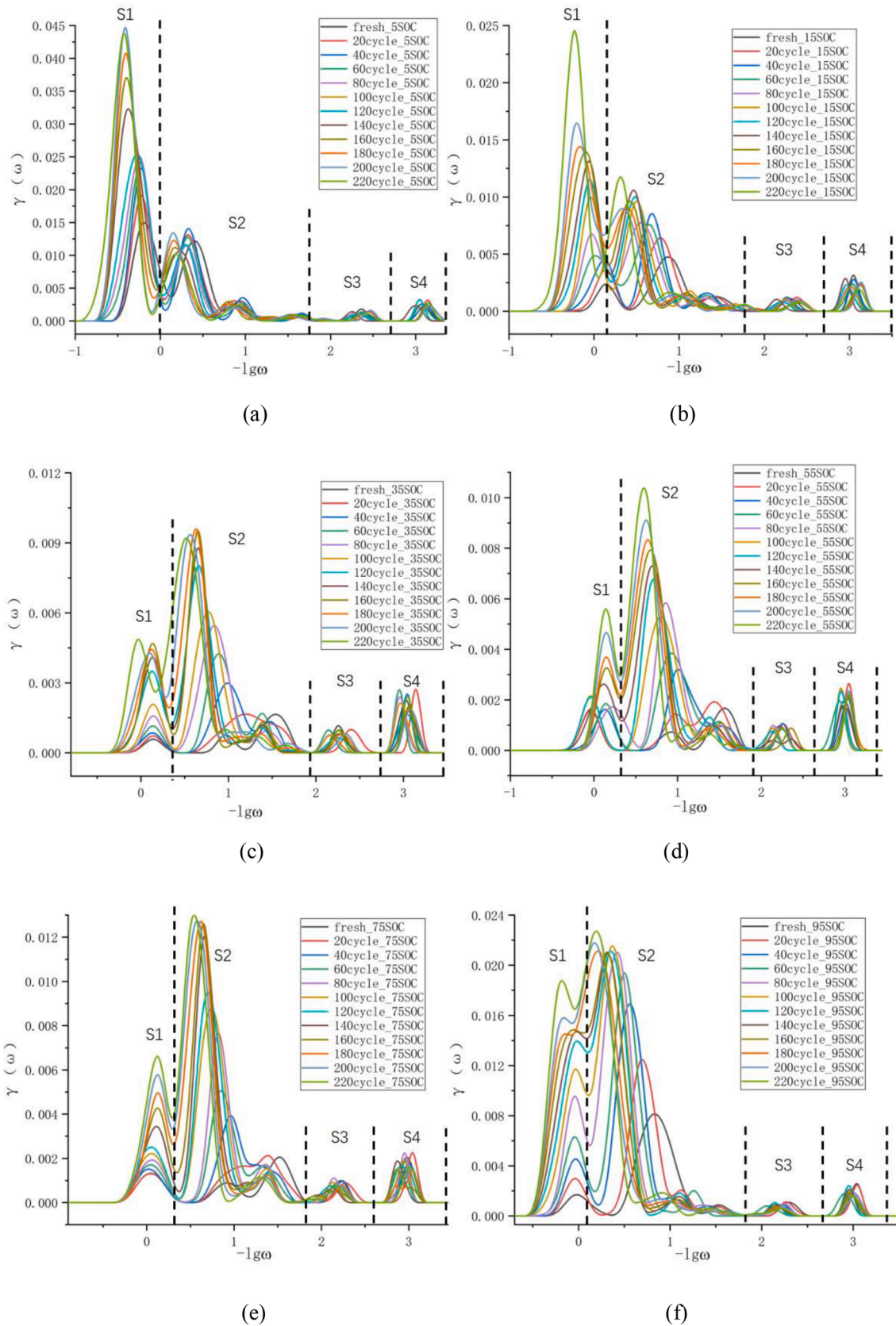


Fig. 4. DRT diagrams of different cycles under different SOC (a) 5%SOC (b) 15%SOC (c) 35%SOC (d) 55%SOC (e) 75%SOC (f) 95%SOC.

Table 2
Governing equations of Electrochemical-thermal model.

Process	Governing equation
Basic process	$U_i = E_{ocv} - \phi_a - \eta_{con} - \eta_{ohm}$ (2)
Liquid phase Ohm's law	$\begin{cases} \kappa_i^{eff} \frac{\partial \phi_{e,i}}{\partial x} = \frac{2RT\kappa_i^{eff}}{F} (t_+ - 1) \frac{\partial \ln c_{e,i}}{\partial x} - i_{e,i}, i = n, sep, p \\ \frac{\partial \phi_{e,n}}{\partial x} \Big _{x=0} = \frac{\partial \phi_{e,p}}{\partial x} \Big _{x=l_n+l_{sep}+l_p} = 0 \end{cases} \quad (3)$
Ohmic polarization	$\begin{cases} \kappa_n^{eff} \frac{\partial \phi_{e,n}}{\partial x} \Big _{x=l_n^-} = \kappa_{sep}^{eff} \frac{\partial \phi_{e,sep}}{\partial x} \Big _{x=l_n^+}, \kappa_{sep}^{eff} \frac{\partial \phi_{e,sep}}{\partial x} \Big _{x=l_n+l_{sep}^-} = \kappa_p^{eff} \frac{\partial \phi_{e,p}}{\partial x} \Big _{x=l_n+l_{sep}^+} \\ \eta_{ohm} = IR_i^{ohm}, i = \text{charge, discharge} \end{cases} \quad (4)$
Solid phase diffusion	$\begin{cases} \frac{\partial c_{s,i}}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_{s,i} r^2 \frac{\partial c_{s,i}}{\partial r} \right), i = n, p \\ D_{s,i} \frac{\partial c_{s,i}}{\partial r} \Big _{r=0} = 0, i = n, p, D_{s,i} \frac{\partial c_{s,i}}{\partial r} \Big _{r=R_i} = -j_i, i = n, p \end{cases} \quad (5)$
Liquid phase diffusion	$\begin{cases} \varepsilon_{e,i} \frac{\partial c_{e,i}}{\partial t} = D_{e,i}^{eff} \frac{\partial^2 c_{e,i}}{\partial x^2} + a_i (1 - t_+) j_i, i = n, p \\ \varepsilon_{e,sep} \frac{\partial c_{e,sep}}{\partial t} = D_{e,sep}^{eff} \frac{\partial^2 c_{e,sep}}{\partial x^2} \\ D_{e,n}^{eff} \frac{\partial c_{e,n}}{\partial x} \Big _{x=l_n^-} = D_{e,sep}^{eff} \frac{\partial c_{e,sep}}{\partial x} \Big _{x=l_n^+} \\ D_{e,sep}^{eff} \frac{\partial c_{e,sep}}{\partial x} \Big _{x=l_n+l_{sep}^-} = D_{e,p}^{eff} \frac{\partial c_{e,p}}{\partial x} \Big _{x=l_n+l_{sep}^+} \end{cases} \quad (6)$
Open circuit voltage	$\begin{cases} c_{e,sep} \Big _{x=l_n+l_{sep}^-} = c_{e,p} \Big _{x=l_n+l_{sep}^+}, c_{e,n} \Big _{x=l_n^-} = c_{e,sep} \Big _{x=l_n^+} \\ E_{ocv} = E_p(x_p^{surf}) - E_n(x_n^{surf}) \\ x_n^{surf} = \frac{c_{s,n}^{surf}}{c_{s,n}^{max}}, x_p^{surf} = \frac{c_{s,p}^{surf}}{c_{s,p}^{max}} \end{cases} \quad (8)$
Electric double layer voltage	$\begin{cases} \phi_i = \phi_{a,i} - \varphi_{1,i}, i = n, p \quad (9) \\ \varphi_{1,i} = \begin{cases} -\frac{2RT_{inner}}{zF} \ln \left(\frac{1}{C_H} \sqrt{2RT_{inner} \varepsilon_0 \varepsilon_r} \right) + \frac{2RT_{inner}}{zF} \ln(\phi_{a,i} - \varphi_{1,i}) - \frac{RT_{inner}}{zF} \ln \sqrt{c_{e,dl,i}}, \varphi_{1,i} > 0 \\ -\frac{2RT_{inner}}{zF} \ln \left(\frac{1}{C_H} \sqrt{2RT_{inner} \varepsilon_0 \varepsilon_r} \right) - \frac{2RT_{inner}}{zF} \ln(\varphi_{1,i} - \phi_{a,i}) + \frac{RT_{inner}}{zF} \ln \sqrt{c_{e,dl,i}}, \varphi_{1,i} < 0 \end{cases} \quad (10) \\ \phi_i = \frac{2RT_{inner}}{F} \ln \left(\sqrt{\left(\frac{j_i}{2i_{0,i}} \right)^2 + 1} + \frac{j_i}{2i_{0,i}} \right) \quad (11) \end{cases}$
Heat production rate	$\begin{cases} \dot{Q} = \dot{Q}_{rev} + \dot{Q}_{irrev} \\ \dot{Q}_{rev} = -IT_{inner} \frac{dE_{ocv}}{dT} \\ \dot{Q}_{irrev} = I(\phi_a + \eta_{con} + \eta_{ohm}) \end{cases} \quad (12)$
Internal temperature of battery	$T_{inner,k+1} = \left(1 - \frac{\Delta t}{m_1 C_1 R_1} \right) T_{inner,k} + \frac{\Delta t}{m_1 C_1} \left(\dot{Q}_k + \frac{1}{R_1} T_{surf,k} \right) \quad (13)$
Battery surface temperature	$T_{surf,k+1} = \left(1 - \frac{\Delta t}{m_2 C_2 R_1} \right) T_{surf,k} + \frac{hS\Delta t}{m_2 C_2 R_2} T_{inner,k} + \frac{hS\Delta t}{m_2 C_2} T_{amb} - \frac{S\Delta t \varepsilon \delta}{m_2 C_2} (T_{surf,k}^4 - T_{amb}^4) \quad (14)$
Parameter update equations	$\begin{cases} E_{ocv}(T_{inner}) = E_{ocv}^{ref} + (T_{inner} - T_{ref}) \frac{dE_{ocv}}{dT} \quad (15) \\ R_{charge}^{ohm} = \frac{R_{charge,ref}^{ohm}}{1 + \alpha_1 (T_{inner} - T_{ref})} \\ R_{discharge}^{ohm} = \frac{R_{discharge,ref}^{ohm}}{1 + \alpha_2 (T_{inner} - T_{ref})} \\ X(T_{inner}) = X(T_{ref}) \exp \left[A_X \left(\frac{1}{T_{ref}} - \frac{1}{T_{inner}} \right) \right] \quad (17) \\ X = D_{s,n}, D_{s,p}, D_{e,n}^{eff}, D_{e,p}^{eff}, D_{e,sep}^{eff}, k_n, k_p \end{cases}$

the reaction polarization overpotential caused by charge transfer reaction, ignoring the electric double effect caused by the accumulation of charge. The other description only focuses on the electric double layer effect without considering the charge transfer reaction. According to the parameter sensitivity analysis results based on EIS in [Section 2](#), the intermediate frequency region represented by the electric double layer changes significantly with aging. But the governing equations of the above two modeling methods do not include parameters that can directly reflect the aging process. Therefore, in this section, an electric double layer model including parameters that can reflect aging are added to the P2D model.

At first, we describe the electric double layer effect based on the GCS (Gouy-Chapman-Stern) model. As shown in [Fig. 5](#), the GCS model divides the electric double layer into two parts: tight layer and dispersed layer. Two parts of residual charges in the electrode surface and the solution are neatly arranged on both sides of the interface. One part of the charges close to the electrode surface form the tight layer. The

surface through the center of the hydrate ion sphere and with a distance of “d” away from the electrode surface from is also called the Helmholtz plane. The other part of the charges are not arranged close to the surface of the electrode but distributed a little farther away from the electrode surface to form a dispersed layer.

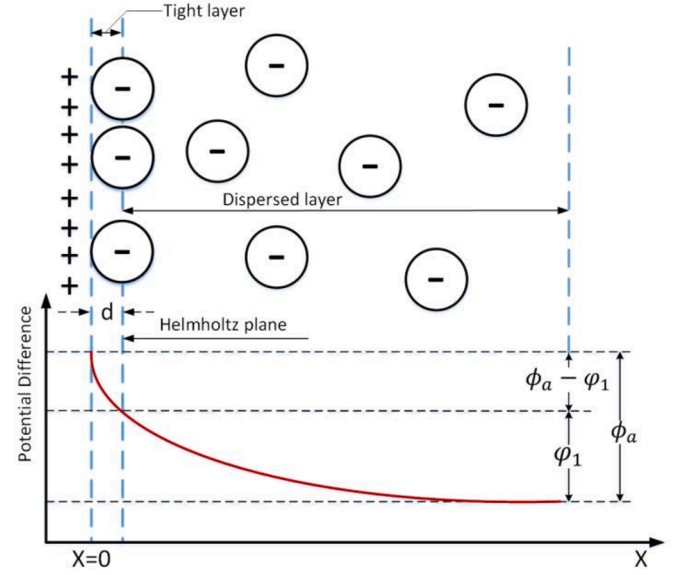
In [Fig. 5](#), ϕ_a indicates the potential difference of the entire electric double layer, $\phi_a - \varphi_1$ indicates the potential difference of the tight layer, and φ_1 indicates the potential difference of the dispersed layer. In the process of solving the electric double layer model, it is assumed that the thickness of the electric double layer is much smaller than the radius of curvature of the electrode surface and each layer parallel to the electrode is itself an equipotential surface. Assuming that the charge is continuously distributed, the Poisson formula of electrostatics can be used in the electric double layer, and the distribution of residual charges in the electric double layer in the electric field obeys Boltzmann's law, and the following relationship can be obtained

Table 3
Model parameters.

Parameters	Description
U_t	Terminal Voltage(V)
E_{ocv}	Open circuit voltage(V)
$\phi_{a,i}$	Total electric double layer potential difference (V)
η_{con}	Concentration polarization overpotential (V)
η_{ohm}	Ohmic polarization overpotential (V)
κ_1^{eff}	Effective electrolyte conductivity ($S\ m^{-1}$)
$\phi_{e,i}$	Liquid phase potential (V)
x	The distance along electrode thickness (m)
$c_{s,i}$	Solid phase lithium ion concentration ($mol\ m^{-3}$)
$c_{e,i}$	Liquid phase lithium ion concentration ($mol\ m^{-3}$)
$\epsilon_{e,i}$	Electrolyte volume fraction (-)
t_+	Liquid phase lithium ion transfer coefficient (-)
$i_{e,i}$	Liquid phase current (A)
a_i	Specific surface area of active particles (m^{-1})
R	Universal gas constant ($J\ mol^{-1}\ K^{-1}$)
F	Faraday constant($C\ mol^{-1}$)
l_i	Electrode or separator thickness (m)
R_i^{ohm}	Ohmic resistance (Ω)
I	External current (A)
$\phi_{1,i}$	Dispersed layer potential difference (V)
ϕ_i	Tight layer potential difference (V)
z	The number of electron transfer in the charge transfer reaction (-)
C_H	Tight layer capacitance (F)
$c_{e,dl,i}$	Lithium ion concentration between liquid phase and the electric double layer ($mol\ m^{-3}$)
ϵ_0	Vacuum dielectric constant (-)
ϵ_r	Relative permittivity (-)
$i_{0,i}$	Exchange current density ($A\ m^{-2}$)
β_i	Symmetry factor (-)
k_i	Reaction rate constant ($m^{2.5}\ mol^{-0.5}\ s^{-1}$)
$D_{s,i}$	Solid phase lithium ion diffusion coefficient ($m^2\ s^{-1}$)
$D_{e,i}^{eff}$	Effective electrolyte lithium ion diffusion coefficient ($m^2\ s^{-1}$)
r	The distance along particle radius (m)
R_i	Particle radius (m)
x_{surf}^i	Surface lithium insertion rate (-)
$c_{s,i}^{max}$	Maximum concentration of active particles ($mol\ m^{-3}$)
$c_{s,i}^{surf}$	Surface concentration of active particles ($mol\ m^{-3}$)
m_1	Electrode winding body quality (kg)
C_1	Specific heat capacity of electrode winding body ($J\ kg^{-3}\ K^{-1}$)
R_1	Thermal resistance between electrode winding body and battery shell (Ω)
m_2	Quality of battery shell(kg)
C_2	Specific heat capacity of battery shell ($J\ kg^{-3}\ K^{-1}$)
R_2	Thermal resistance between electrode battery shell and ambience (Ω)
h	Heat transfer coefficient between battery shell and environment ($W\ m^{-2}\ K^{-1}$)
S	Heat exchange area between battery shell and environment (m^2)
ϵ	Emissivity (-)
δ	Blackbody radiation constant ($W\ m^{-2}\ K^{-4}$)
E_{ocv}^{ref}	Reference open circuit voltage(V)
$\dot{Q}$	Heat production rate (W)
$\dot{Q}_{rev}$	Reversible heat production rate (W)
$\dot{Q}_{irrev}$	Irreversible heat production rate (W)
T_{amb}	Ambient temperature (K)
T_{inner}	Battery Internal temperature (K)
T_{surf}	Battery surface temperature (K)
T_{ref}	Reference temperature (K)
$R_{charge,ref}^{ohm}$	Charging ohmic resistance reference value (Ω)
R_{charge}^{ohm}	Charging ohmic resistance (Ω)
$R_{discharge,ref}^{ohm}$	Discharging ohmic resistance reference value (Ω)
$R_{discharge}^{ohm}$	Discharging ohmic resistance (Ω)
	Temperature coefficient of charging ohmic resistance (K^{-1})

Table 3 (continued)

Parameters	Description
α_1	
α_2	Temperature coefficient of discharging ohmic resistance (K^{-1})
$X(T_{inner})$	The value of parameter X at T_{inner} (-)
$X(T_{ref})$	The value of parameter X at T_{ref} (-)
A_X	Pre-factor of parameter X (-)

**Fig. 5.** GCS electric double layer model and interface potential distribution.

$$\frac{d\phi_x}{dx} = -\sqrt{\frac{2RTc}{\epsilon_0\epsilon_r}} \left[\exp\left(\frac{F\phi_x}{2RT}\right) - \exp\left(-\frac{F\phi_x}{2RT}\right) \right] \quad (20)$$

where ϕ_x is the potential at the position with the distance x away from the electrode surface (V).

According to Gauss's theorem, the relationship between the surface charge density of the electrode and the potential gradient near the surface can be obtained

$$\sigma = \epsilon_0\epsilon_r \left(\frac{d\phi_x}{dx} \right)_{x=0} = -\epsilon_0\epsilon_r \left(\frac{d\phi_x}{dx} \right)_{x=d} \quad (21)$$

where σ is the electrode surface charge density ($C\cdot m^{-2}$).

Assuming that the tight layer capacitance C_H is a constant value, the surface charge density of the electrode is used as the intermediate amount to obtain the relationship between the tight layer potential, the dispersed layer potential, and the total potential of the electric double layer. Combining the Eq. (20) and Eq. (21), we can obtain:

$$\phi_a = \phi_1 + \frac{\sigma}{C_H} = \phi_1 + \frac{1}{C_H} \sqrt{2RTc\epsilon_0\epsilon_r} \left[\exp\left(\frac{zF\phi_x}{2RT}\right) - \exp\left(-\frac{zF\phi_x}{2RT}\right) \right] \quad (22)$$

When the energy of the coulomb force acting between the residual charges and the electrodes in the electric double layer is much greater than the energy of the thermal movement of the residual charges, the item $\exp\left(-\frac{zF\phi_x}{2RT}\right)$ in Eq. (22) can be ignored and Eq. (22) can be simplified as Eq. (10)

The electric double layer only describes the accumulation of residual charge, the charge transfer should also be modeled. Assuming that the charge transfer reaction only take place in the tight layer of the electric double layer, we describe the charge transfer reaction using the B-V

equation as:

$$j_i = i_{0,i} \left[\exp \left(\frac{(1 - \beta_i)F}{RT} \phi_i \right) - \exp \left(\frac{\beta_i F}{RT} \phi_i \right) \right], i = n, p \quad (23)$$

$$i_{0,i} = k_i \left(c_{s,i}^{\max} - c_{s,i}^{\text{surf}} \right)^{1-\beta_i} \left(c_{s,i}^{\text{surf}} \right)^{\beta_i} c_e^{1-\beta_i}, i = n, p \quad (24)$$

When the symmetry factor β_i , $i = n, p$ is 0.5, Eq. (24) can be expressed by the hyperbolic sine function as shown below

$$j_i = 2i_{0,i} \sinh \left(\frac{0.5F}{RT} \phi_i \right), i = n, p \quad (25)$$

By taking the inverse function of Eq. (25), the overpotential of tight layer of the electric double layer can be solved, and the tight layer potential is described as Eq. (11).

Combining Eq. (9), Eq. (10) and Eq. (11), the potential difference of the electric double layer can be obtained.

3.2. Thermal coupling model

The establishment of the thermal model will be based on the following three assumptions:

- (1) Ignore the unevenness of the internal temperature distribution of the battery in the radial direction.
- (2) Ignore the unevenness of the internal temperature distribution in the axial direction of the battery.
- (3) Ignore the temperature transfer in the thickness direction of the battery plate.

The heat production rate of the battery can be divided into polarization heat, ohmic heat, and reaction heat. The reaction heat is reversible heat, which manifests as endothermic or exothermic heat during charging and discharging, while polarization heat and ohmic heat are collectively referred to as irreversible heat, which manifests as exothermic heat during charging and discharging. Hence the heat production rate is described as the sum of the reversible heat production rate and the irreversible heat production as shown in Eq. (12). $\frac{dE_{ocv}}{dT}$ in Eq. (12) is entropy coefficient ($V \cdot K^{-1}$), which can be measured in the same way as our previous work [33].

Fig. 6 shows the radial concentrated mass thermal model. The electrode winding body and the battery casing are regarded as the research objects respectively, the differential equations of heat conduction between the electrode winding body and the battery casing, and between the battery casing and the environment can be obtained from the law of conservation of energy. Their discretized equations are shown as Eq. (13) and Eq. (14), which are used for real-time simulation of battery internal temperature and surface temperature.

Regarding the parameter update with temperature, the form of Nernst equation is used to update E_{ocv} with temperature in real time through entropy coefficients shown in Eq. (15), the form of updating the charge and discharge ohmic internal resistance is described as Eq. (16) and the remaining 7 parameters are updated using Arrhenius formula as

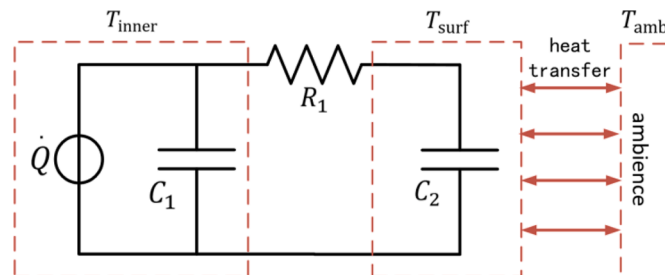


Fig. 6. Radial concentrated mass thermal model of battery.

Eq. (17).

4. Model parameter acquisition

4.1. Parameter sensitivity analysis based on terminal voltage response

This section does not study the sensitivity of battery parameters to the aging state, but only considers the influence degree of the parameters on the terminal voltage response of the battery under certain number of cycles. The parameter sensitivity analysis compares variation of terminal voltages of battery models with different parameters sets. The parameter set contains 15 parameters as shown in Fig. 7, all these parameters are assigned to 6 different values. A parameter is regarded as the highly sensitive parameter if the terminal voltages of battery models in 6 different values of this parameter show a significant variation. Otherwise, parameters are regarded as the weakly sensitive parameters.

The parameter sensitivity analysis is carried out in different conditions. According to working ranges of batteries, two constant current discharging conditions with the current rates of 0.1C and 1C are chosen to analyze the parameter sensitivity. Both two condition are carried out with 25°C ambient temperature and a SOC range from 10% to 100%. The average variation of the terminal voltage U_t of each parameter in 100%SOC is shown in Fig. 7.

It can be seen from the simulation results that $D_{s,i}$, R_i^{ohm} , $c_{s,i}^{\max}$, C_H , k_i and $\epsilon_{s,i}$ are highly sensitive parameters. These group of parameters should be paid attention to during parameter identification. And $\epsilon_{e,i}$, $D_{e,i}^{\text{eff}}$, κ_e^{eff} are weakly sensitive parameters, and their values only need to limit to reasonable ranges during parameter identification.

4.2. Parameter acquisition

The proposed electrochemical-thermal model contains many parameters, and the single parameter acquisition method cannot accurately obtain the value of each parameter. Different parameter acquisition methods according to the type of the parameters are used in this section.

The parameters in the electrochemical-thermal model are divided into three groups. The first group of parameters are directly measured by disassembling the battery or making coin cells with the same chemistry as NCR18650BD battery, including l_p , l_n , l_{sep} , $A_{0,p}$, $A_{0,n}$, x_0^n , x_0^p and OCV curves. The second group of parameters are obtained from some references, including h , R_1 , m_1 , C_1 , ϵ and δ . The third group of parameters is identified using the genetic algorithm, including $D_{s,p}$, $D_{s,n}$, D_e , $\epsilon_{s,p}$, $\epsilon_{s,n}$,

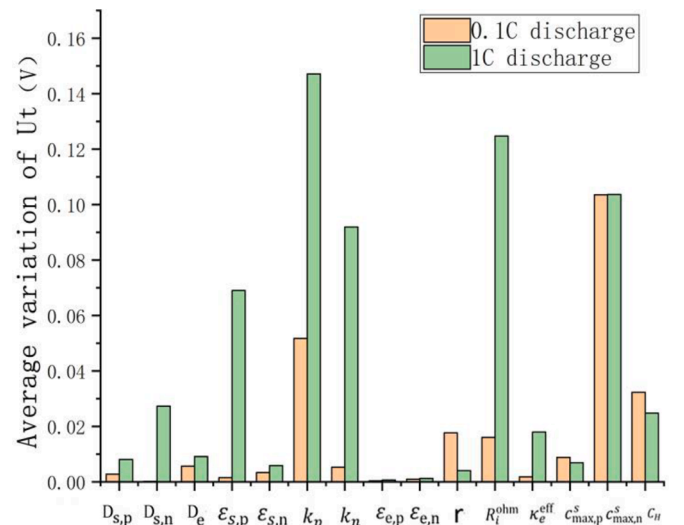


Fig. 7. The average U_t variation of each parameter on the full SOC.

$k_p, k_n, \epsilon_{e,p}, \epsilon_{e,n}, r, R_{discharge}^{ohm}, R_{charge}^{ohm}, c_{max,p}^s, c_{max,n}^s, \kappa_e^{eff}, C_H, m_2, C_2, \alpha_1, \alpha_2$ and $A_X, X = D_{s,n}, D_{s,p}, D_{e,n}^{eff}, D_{e,p}^{eff}, D_{e,sep}^{eff}, k_n, k_p$.

4.2.1. The first and second groups parameters acquisition

In the first groups, positive and negative electrode plate thickness, separator thickness, and positive and negative electrode plate area of NCR18650BD battery are directly measured by disassembling the battery. The results are shown in Table 4.

Because materials of the positive electrode and the negative electrode of NCR18650BD battery are lithium cobalt oxides (LCO) and graphite, respectively, two kinds of coin cells, including LCO-Li cell and graphite-Li cell, are made to obtain the x_0^n, x_0^p and OCV curves. The OCV curves of these two kinds of coin cells and whole NCR18650 cells are measured using charging and discharging current of 1/40C. We fit OCV-lithium insertion rate curves of two kinds of coin cells and subtract the curve of LCO-Li cell with the curve of graphite-Li cell to fit the measured OCV-SOC curve of the whole NCR18650 cell. Taking the error between the superposed OCV curve and the measured OCV curve as the objective function, the least square method is used to identify the initial lithium insertion rate x_0^n of the negative electrode, and the initial lithium insertion rate x_0^p of the positive electrode. Identified x_0^n and x_0^p are 0.8937 and 0.3904, respectively and the fitted OCV- lithium insertion rate curves of coin cells are as follows:

$$E_n = 0.74086 + 7.40322 * u - 20.06007 * u^2 + 55.40734 * u^3 - 94.9406 * u^4 + 85.32438 * u^5 - 30.59186 * u^6 - 3.53292 * u^{0.5} \quad (26)$$

$$E_p = 1906.873 + 20,013.69881 * u - 28,834.50163 * u^2 + 39,321.32318 * u^3 - 35,429.03 * u^4 + 18,305.237 * u^5 - 4096.11748 * u^6 - 11,186.366 * u^{0.5} \quad (27)$$

where Eq. (26) is fitted OCV- lithium insertion rate curves of graphite-Li cell and Eq. (27) is fitted OCV- lithium insertion rate curves of LCO-Li cell. The fitted and measured OCV curve are shown in Fig. 8, and the root mean square error (RMSE) of the fitting result is 0.013 V.

Regarding parameters in the second group, their values and the references are shown in Table 5.

4.2.2. The third group parameters acquisition

Parameters in the third group can also be divided into three parts: electrochemical model parameter including $D_{s,p}, D_{s,n}, D_e, \epsilon_{s,p}, \epsilon_{s,n}, k_p, k_n, \epsilon_{e,p}, \epsilon_{e,n}, r, R_{discharge}^{ohm}, R_{charge}^{ohm}, c_{max,p}^s, c_{max,n}^s, \kappa_e^{eff}, C_H$, thermal model parameters including m_2, C_2 and parameters used for parameter update including $\alpha_1, \alpha_2, A_X, X = D_{s,n}, D_{s,p}, D_{e,n}^{eff}, D_{e,p}^{eff}, D_{e,sep}^{eff}, k_n, k_p$.

MSCC (Multi-Stage Constant Current) condition is introduced into electrochemical model parameter identification. The condition is a custom mixed constant current pulse working condition. As shown in Fig. 9(a), the condition includes 1.5C charging/discharging current and 3C discharging current. The charging/discharge current rates are determined with reference to the battery manual. Because the frequency range corresponding to different polarization processes are different, the frequency distribution range of the current excitation should be wider. According to the EIS analysis in Section 2.2, the general ranges of time constants corresponding to different polarization processes can be

obtained. Accordingly, current pulses in MSCC condition are designed with different charging and discharging time, including 6 s, 60 s, and 600 s and enough resting times are set. This custom condition can decouple different polarization processes and is suitable for parameter identification.

The terminal voltage curve of the fresh cell under MSCC conditions at 25 °C is shown in Fig. 9(b). The influences on the shape of the terminal voltage curve are explored by separately changing the values of different parameters. Different color regions correspond to different parameters. The starting point of voltage curve marked by gray arrow can reflect the initial SOC of the battery, and the initial lithium insertion rate of the positive electrode and negative electrode can be identified based on the point. The terminal voltage in the blue circle region can reflect the real-time SOC. Combining the fitted OCV curve as shown in Eq. (26) and Eq. (27), the maximum solid phase concentration $c_{max,i}^s$ of the battery can be identified based on identified SOC. The terminal voltage curve in the purple circle is highly related to the D_e . The region marked in the pink region corresponds to the concentration polarization, which can be used to guide the identification of $D_{s,p}, D_{s,n}$ and D_e . The terminal voltage characteristic of the low SOC in the green region is characterized by a "valley" phenomenon, which is caused by the electric double layer. Combined with the pink region, it can be used to identify the k_n, k_p and C_H .

Based on above analysis, model parameters can be identified by a stimulus-response method, which obtains parameter values by adjusting parameter to fit the measured voltage curve. The values of some battery parameters in the beginning of life have been determined by the method, and the value ranges of some parameters are narrowed, as shown in Table 6, which is also the preparation for the subsequent parameter identification using the genetic algorithm (GA).

GA is one of common methods to identify parameters, which is based on Darwin's theory of evolution and can search the optimal solution by simulating natural selection and biological genetic mechanisms. This optimization algorithm does not have the restriction of function derivability and function continuity, has strong global search ability, and can adjust the search direction of the optimal solution by itself.

The subsequent parameter identification using GA includes three steps: The electrochemical model parameters, not identified by the stimulus-response method, are identified using GA in the first step. In this step, the simulation is carried out only based on the electrochemical model, which aims to simulate the voltage response of the measured battery in MSCC condition and the objective function is expressed as Eq. (28). The second step introduce the thermal model into the electrochemical model with identified electrochemical parameters to identify the thermal model parameters based on the surface temperature of the battery in 0.4C constant current discharge condition and the objective function in this step is expressed as Eq. (29). After electrochemical model parameters and thermal model parameters, the coupling relationship of electrochemical model parameters and temperature is introduced to build electrochemical-thermal model, and parameters used in parameter update are identified in the third step.

$$\min f = a * \sqrt{\frac{\sum_{i=0}^n (U_{t,i} - \widehat{U}_{t,i})^2}{n}} + b * \max |U_{t,i} - \widehat{U}_{t,i}| \quad (28)$$

$$\min f = a * \sqrt{\frac{\sum_{i=0}^n (T_{surf,i} - \widehat{T}_{surf,i})^2}{n}} + b * \max |T_{surf,i} - \widehat{T}_{surf,i}| \quad (29)$$

where $U_{t,i}$ is the measured terminal voltage, $\widehat{U}_{t,i}$ is the simulated terminal voltage, $T_{surf,i}$ is the measured surface temperature, $\widehat{T}_{surf,i}$ is the simulated surface temperature, n is the length of data, a is weight of RMSE and b is weight of maximum error (ME).

Above parameter identification process is validated based on the data of the fresh cell in condition at 25 °C. The simulated terminal

Table 4
Electrode size parameters.

Parameters	Values	Units
l_n	0.0789	mm
l_p	0.0483	mm
l_{sep}	0.0236	mm
A_n	68,503.2	mm ²
A_p	65,992	mm ²

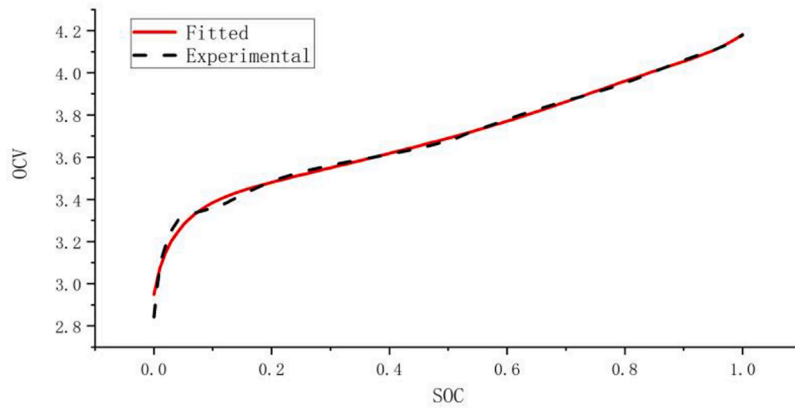


Fig. 8. The OCV curve of NCR18650BD battery at 25 °C.

Table 5
Parameters in the second group.

Parameters	Values	Units	Reference
h	20	$\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$	[34]
R_1	1.7732	$\text{K}\cdot\text{W}^{-1}$	[35]
m_1	0.02671	kg	[35]
C_1	998	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	[35]
ε	0.04	–	[34]
δ	5.67×10^{-8}	$\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$	[34]

voltages in the first step and the third step are compared in Fig. 9(c), RMSE between simulated voltages and measured voltages in the first step is 0.01177 V and ME is 0.041977 V. After introducing the coupling relationship of electrochemical model parameters and temperature, RMSE and ME in the third step respectively reduce by 0.00227 V and 0.00459 V, respectively. The validation of simulated surface temperature in the second step is shown in Fig. 10, and RMSE and ME between simulated temperatures and measured temperatures are 0.051968551 K and 0.15905 K, respectively.

5. Model validation and SOH estimation

The change of MSCC curve with aging is shown in Fig. 9(d). The OCV

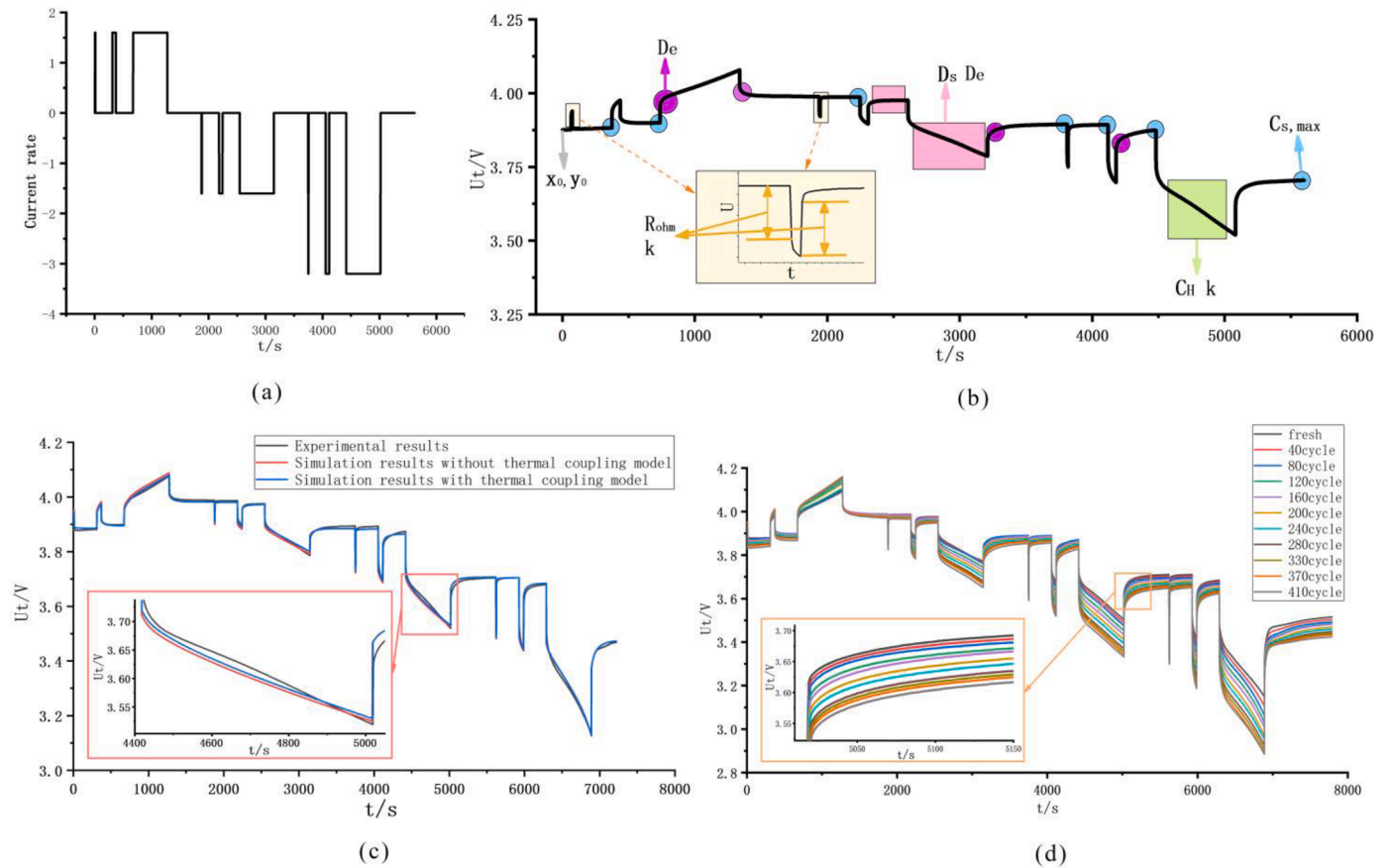
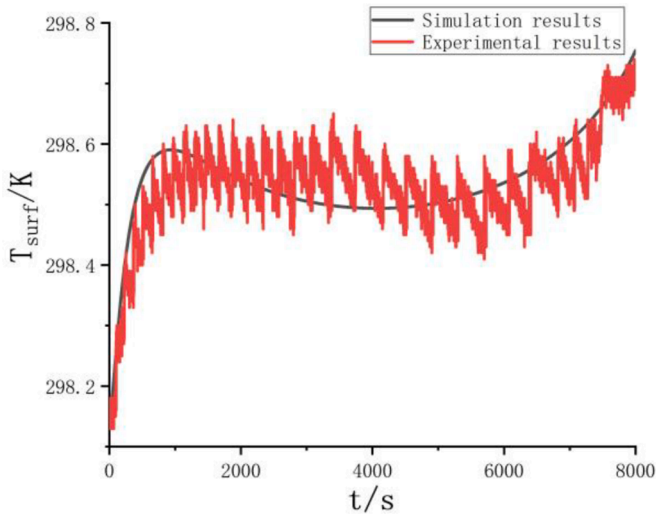


Fig. 9. (a) MSCC current rate (b) Terminal voltage curve under MSCC (c) Terminal voltage comparison between different models (d) MSCC curves under different cycles.

Table 6

Certain values or value ranges of some parameters.

Parameters	Values	Units
x_0	0.8848	–
y_0	0.3954	–
$c_{\max,p}^s$	140,378.6	$\text{mol} \cdot \text{m}^{-3}$
$c_{\max,n}^s$	47,591.1	$\text{mol} \cdot \text{m}^{-3}$
R	2.8×10^{-7}	M
D_e	$3 \sim 5 \times 10^{-11}$	$\text{mol} \cdot \text{m}^{-3}$
$D_{s,i}$	$2 \sim 4 \times 10^{-15}$	$\text{mol} \cdot \text{m}^{-3}$
$R_{\text{ohm}}^{\text{charge}}$	0.005~0.015	Ω
$R_{\text{ohm}}^{\text{discharge}}$	0.012~0.016	Ω
k_p	$4 \sim 8 \times 10^{-12}$	$\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$
k_n	$4 \sim 8 \times 10^{-11}$	$\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$
C_H	0.4~0.6	$\text{F} \cdot \text{m}^{-2}$

**Fig. 10.** Battery surface temperature comparison between measured and simulated results.

in the same SOC decreases slightly as the number of cycles increases. When the same amount of capacity is consumed, the SOC of the battery continues to decrease, which shows that the capacity of the battery is constantly decreasing. At the end of each pulse, the arc of the terminal voltage curve keeps increasing, which shows that the time constant of the diffusion part is increasing. The "valley" phenomenon of the last pulse is gradually obvious, which is related to the shape change of the OCV curve at low SOC and the change of the electric double layer voltage. The change trend of MSCC curve has certain guiding significance for parameter acquisition under different cycles.

According to the parameter acquisition method introduced in Section 4, within the range of 0 to 450 cycles, the experimental values of MSCC conditions are used to obtain a set of electrochemical parameter values every certain number of cycles. RMSE and ME between simulated terminal voltages and measured terminal voltages under some cycles are shown in Table 7.

To further verify the accuracy of the model parameters under actual road conditions, CLTC-P standard road conditions are used for verification. Fig. 11(a) shows the current of CLTC-P conditions, which is calculated by introducing the velocity-time spectrum of CLTC-P condition into the vehicle dynamic model. The parameters of the electrochemical-thermal model are verified under the 0th cycle, 280th cycle, and 450th cycle aging stages. Introducing the parameter values identified by MSCC conditions under these three cycles into the electrochemical-thermal model, the accuracy of the model in CLTC-P

Table 7

RMSE and MR of terminal voltages in MSCC condition from 0 to 450 cycle.

Cycle number	RMSE (V)	MR (V)
0	0.002405	0.041977
40	0.003743	0.047756
80	0.002336	0.046922
120	0.002432	0.042516
160	0.002442	0.049025
200	0.002505	0.049883
240	0.002368	0.048352
280	0.003009	0.048310
330	0.002958	0.047789
370	0.003176	0.048433
410	0.002651	0.053122
450	0.002730	0.055658

conditions is validated. Fig. 7(b)(c)(d) show the comparison between the measured and simulated terminal voltage values of the CLTC-P conditions under the above three cycles. RMSEs are 0.0168 V, 0.0225 V, and 0.0254 V respectively.

5.1. SOH estimation

The discharge capacity tests at 25 °C are carried out every certain number of cycles and the discharge capacities are used to calculate the actual SOH of the battery. The calculation formula is as follows

$$\text{SOH} = C / C_0 \quad (30)$$

where C is current battery discharge capacity, C_0 is fresh battery discharge capacity.

The discharge capacity under different cycles and the calculated SOH under the corresponding cycle capacity are shown in Table 8.

Among the many electrochemical model parameters identified in aging process, only seven parameters x_0 , y_0 , $c_{\max,p}^s$, $c_{\max,n}^s$, $R_{\text{charge}}^{\text{ohm}}$, $R_{\text{discharge}}^{\text{ohm}}$ and C_H show monotonic trends with the capacity degradation, which are suitable for SOH estimation. As shown in Table 8, SOH shows an almost linear decrease except the large drop in the first 40 cycles, so the Pearson correlation coefficient ρ is introduced to further verify the linear correlation between these seven SOH-related parameters and SOH, ρ is defined as the quotient of the covariance and standard deviation between the two variables X and Y . The Pearson correlation coefficient ρ is a value between 1 and -1 . When $\rho = 1$, it means that X and Y are completely positively correlated; when $\rho = 0$, it means that X and Y are irrelevant; when $\rho = -1$, it means that X and Y are completely negatively correlated.

The calculation method is shown in Eq. (31).

$$\begin{aligned} \rho_{X,Y} &= \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E((X - \mu_X)(Y - \mu_Y))}{\sigma_X \sigma_Y} \\ &= \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - E^2(X)} \sqrt{E(Y^2) - E^2(Y)}} \end{aligned} \quad (31)$$

where variables X is the value of the SOH-related parameter, and Y is the measured values of SOH

The ρ calculation results of the above seven SOH-related parameters are shown in Table 9.

Then the SOH estimation is completed with the help of Back propagation (BP) neural network. BP neural network is a multi-layer feed-forward network trained by error back propagation. BP algorithm uses gradient search technology to minimize the RMSE between the actual output value and the expected output value of the network.

The inputs of the BP neural network are electrochemical parameter values under 12 cycle numbers in Table 8, and the output are SOH values under the same cycle numbers. The SOH estimation results are shown in Fig. 12.

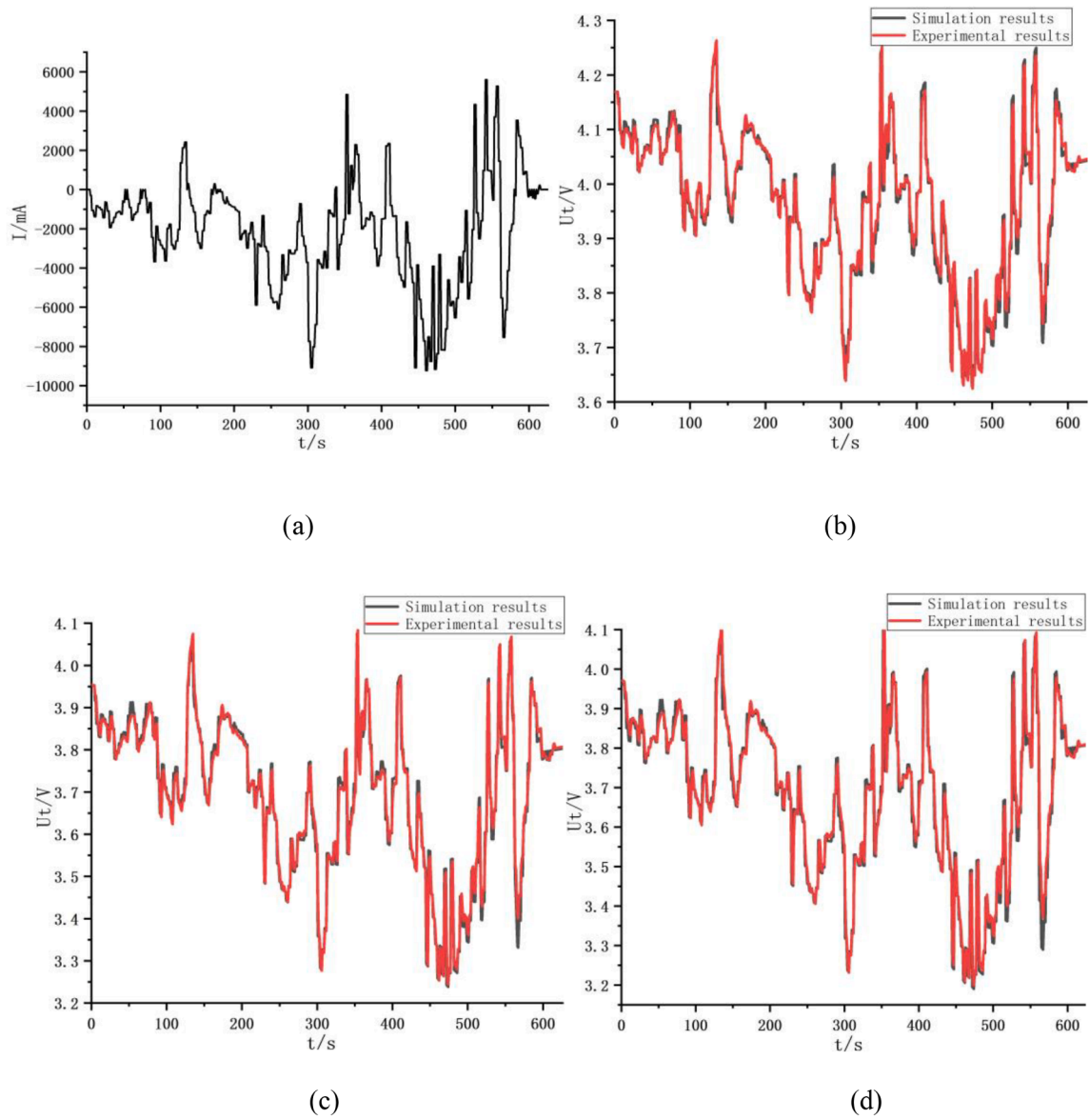


Fig. 11. (a) CLTC-P condition current and terminal voltages of (b) 0th cycle, (c) 280th cycle and (d) 450th cycle in CLTC-P condition.

Table 8

0 to 450cycle battery discharge capacity and SOH.

Cycle number	Discharge capacity (mAh)	SOH
0	3009.304	1
40	2898.836	0.96329
80	2885.902	0.95899
120	2873.513	0.95487
160	2850.548	0.94724
200	2830.723	0.94065
240	2810.900	0.93406
280	2798.004	0.92978
330	2771.631	0.92102
370	2746.215	0.91257
410	2736.509	0.90931
450	2715.891	0.90249

Table 9

Pearson correlation coefficient calculation results.

Parameter	ρ
x_0	0.966886524
y_0	-0.966886524
$c_{\max,p}^s$	0.939046148
$c_{\max,n}^s$	0.841820203
$R_{\text{charge}}^{\text{ohm}}$	-0.940875042
$R_{\text{discharge}}^{\text{ohm}}$	-0.945795571
C_H	-0.948905141

As shown in Fig. 12, ME between the simulated SOH and the actual SOH is less than 1.52%. RMSE between the simulated SOH and the actual SOH is only 0.625%

6. Conclusion

This paper aims to develop an electrochemical-thermal model to describe the lithium battery degradation and achieve SOH estimation. At

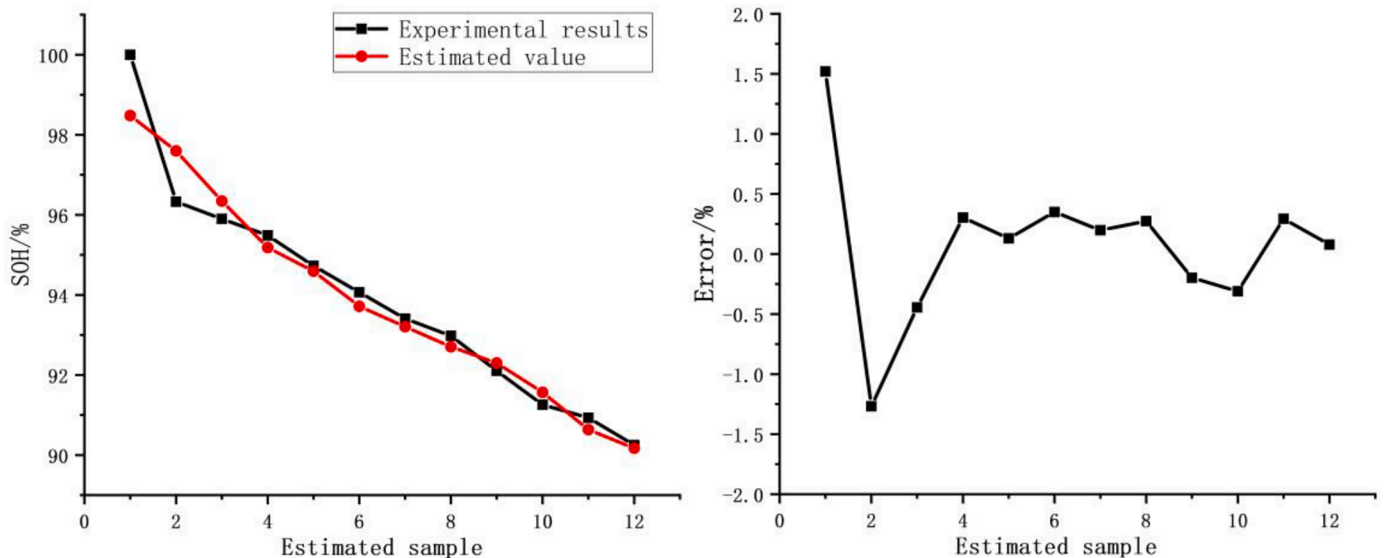


Fig. 12. SOH estimation result.

first, we analyze the impedance change of different electrochemical processes inside lithium ion batteries in the aging process based on EIS and DRT. In this section, we conclude that reaction processes in the electric double layer and the diffusion processes in solid and liquid phase are highly sensitive to the battery degradation. Then, the electrochemical-thermal model is developed, and this model give a detailed description on the charge transfer and charge accumulation in the electric double layer combining GCS model and P2D model. In section of parameter identification, the MSCC condition is designed to decouple external characteristics caused by different internal processes of batteries. Some regions in the terminal voltage curve of MSCC condition are marked to correspond to different electrochemical parameter, based on which a stimulus-response method is used to obtain values or ranges of some parameters. Then a three-step parameter identification method based on GA is proposed to identify all other parameters. In the final section, all parameters are identified based on MSCC condition data in aging experiments and the accuracy of the electrochemical-thermal model is verified in CLTC-P standard road condition in some aging stage. Besides, seven highly SOH-related parameters are found out and an accurate SOH estimation is achieved based on these parameters.

CRediT authorship contribution statement

Dafang Wang: Supervision, Project administration, Funding acquisition, Resources. **Qi Zhang:** Conceptualization, Validation, Writing – review & editing. **Huanqi Huang:** Methodology, Software, Writing – original draft. **Bowen Yang:** Validation, Writing – original draft. **Hao-song Dong:** Writing – review & editing, Software. **Jingming Zhang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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